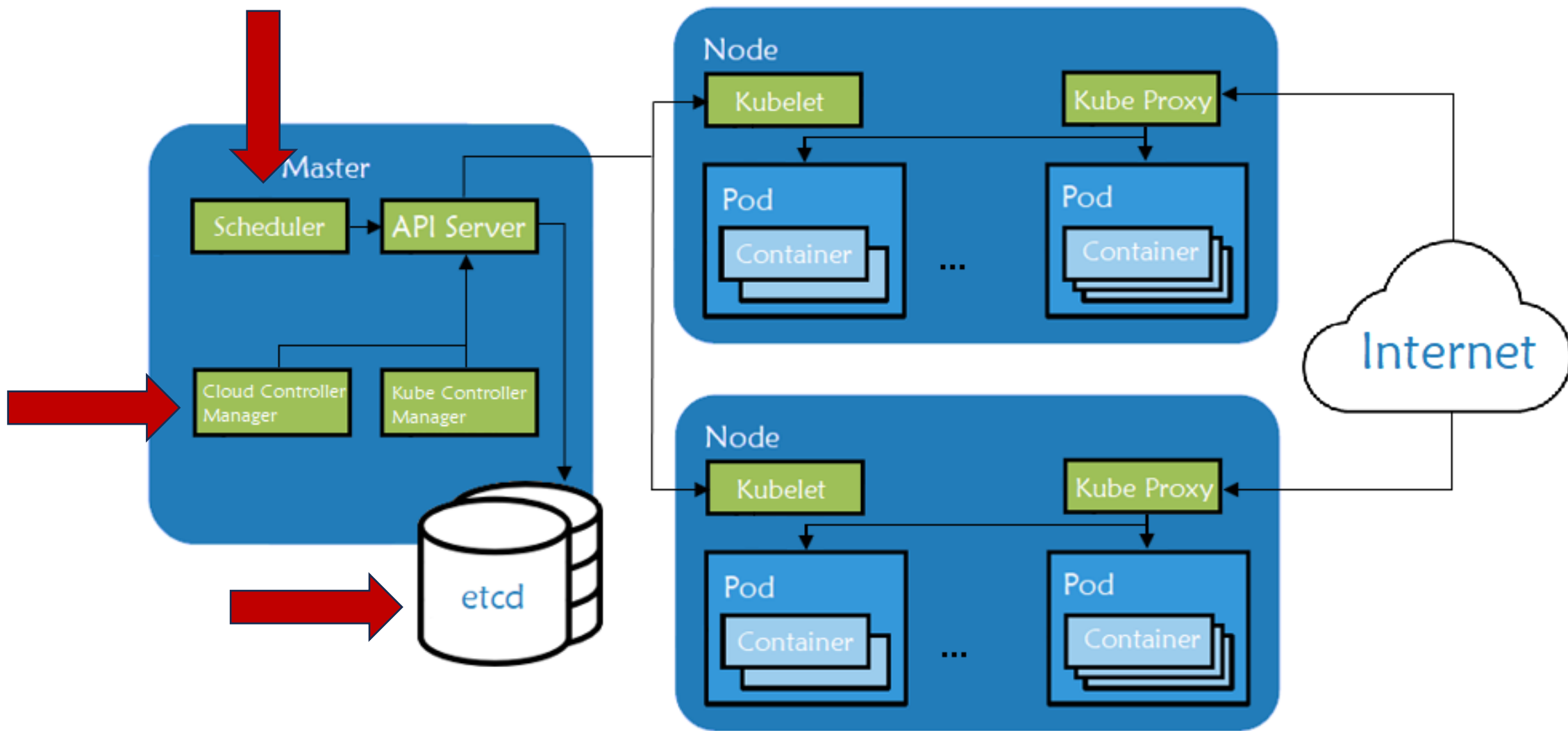


Different Parts in a Master Node of



kubernetes



ETCD



**ETCD is a distributed
reliable key-value store
that is Simple, Secure &
Fast**

key-value store

Key	Value
Name	John Doe
Age	45
Location	New York
Salary	5000

Put Name "John Doe"

Get Name

"John Doe"

Tabular/Relational Databases

Name	Age	Location
John Doe	45	New York
Dave Smith	34	New York
Aryan Kumar	10	New York
Lauren Rob	13	Bangalore
Lily Oliver	15	Bangalore

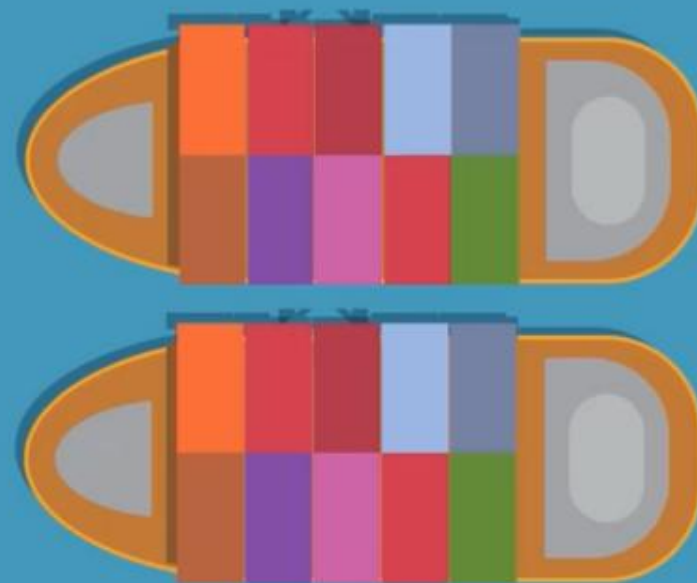


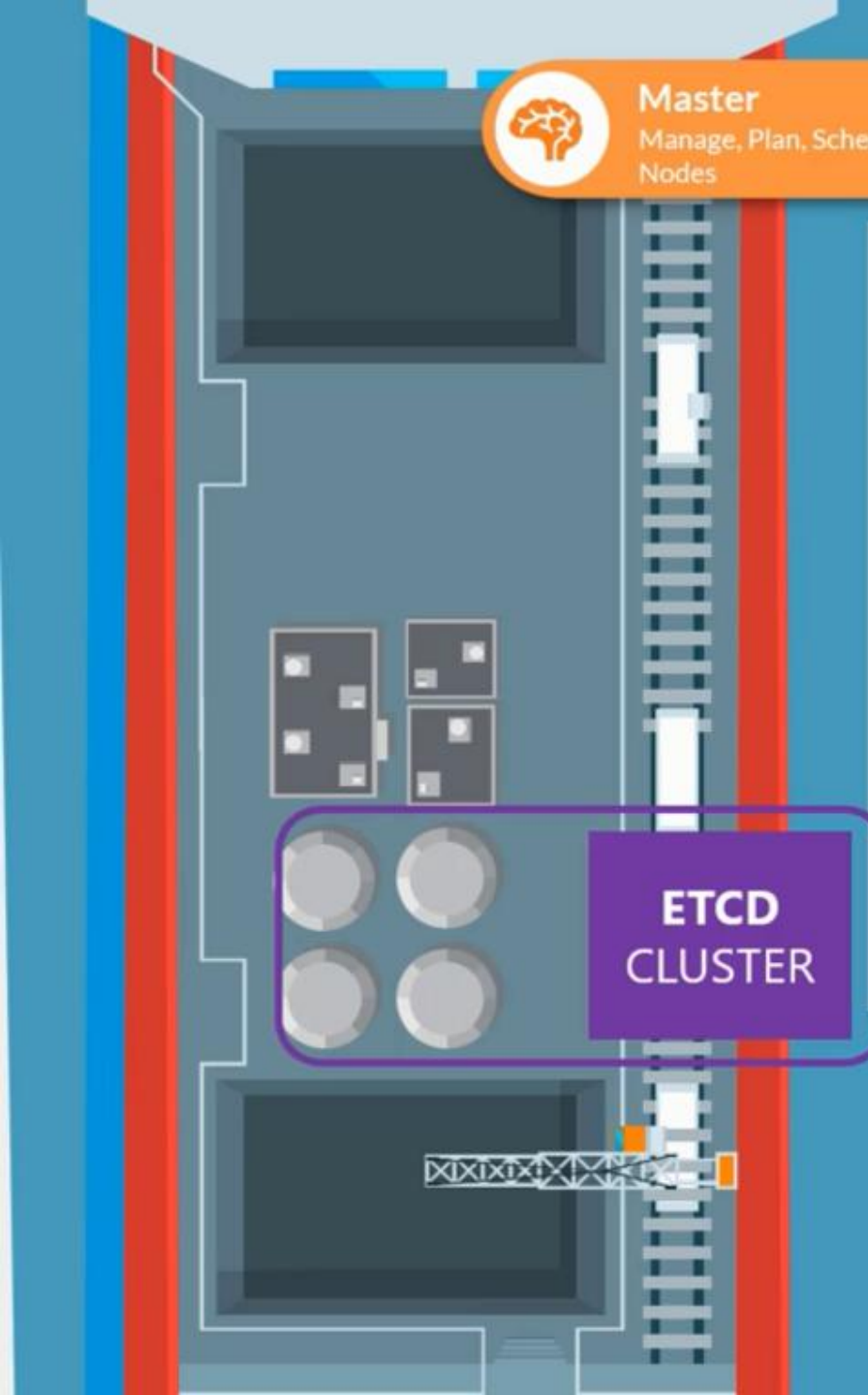
Master

Manage, Plan, Schedule, Monitor
Nodes

What we
store in ETCD?

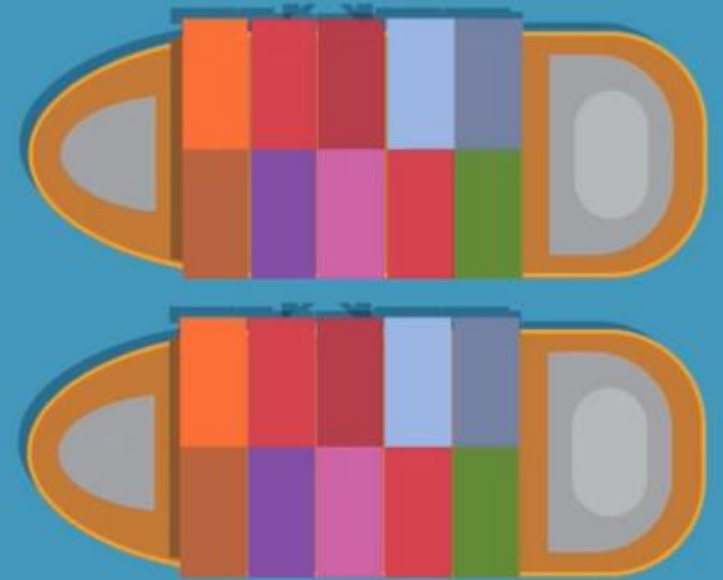
**ETCD
CLUSTER**



A stylized illustration of a control room or bridge. On the left, there are several control panels with buttons and screens. A central vertical console has a ladder-like structure. At the top, an orange pill-shaped box contains a brain icon and the text 'Master Manage, Plan, Schedule, Monitor Nodes'. Below it, a purple box labeled 'ETCD CLUSTER' is connected to the central console. At the bottom, a crane-like structure is visible. The background is a solid blue color.

Master
Manage, Plan, Schedule, Monitor
Nodes

What we do
not store?



1. Container images (stored in container registry)
2. Application data/logs (stored in volumes/external storage)
3. Actual running processes (managed by container runtime)

| ETCD in HA Environment

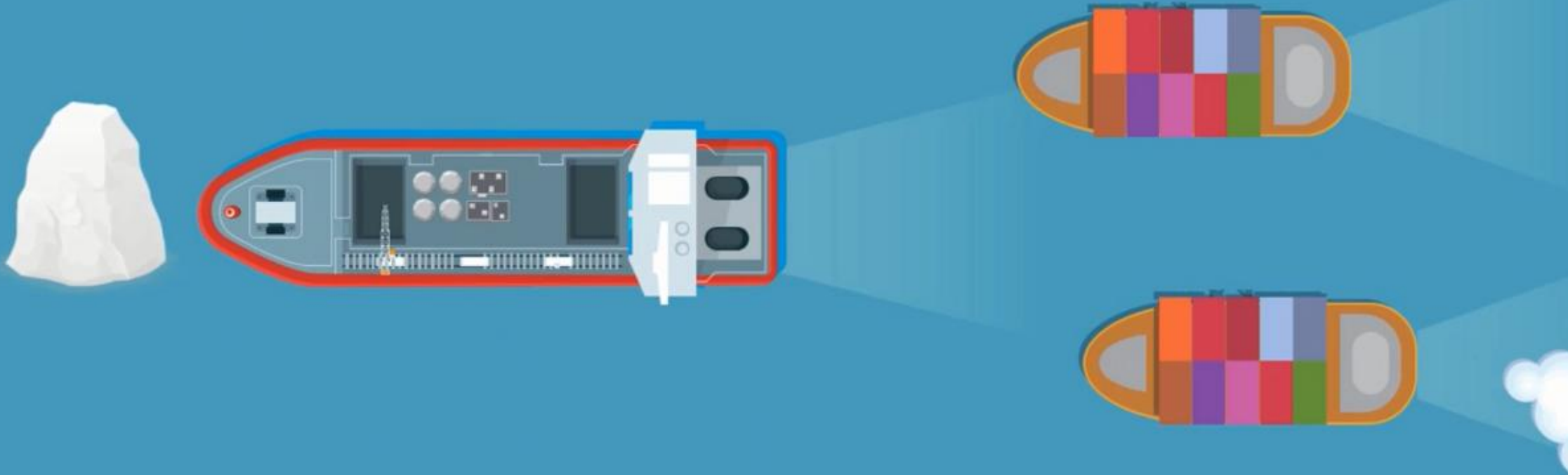
| ETCD in HA Environment



- **etcd** listens on port **2379** for client requests.
- It also uses port **2380** for etcd peer-to-peer communication.

HA for Master Node

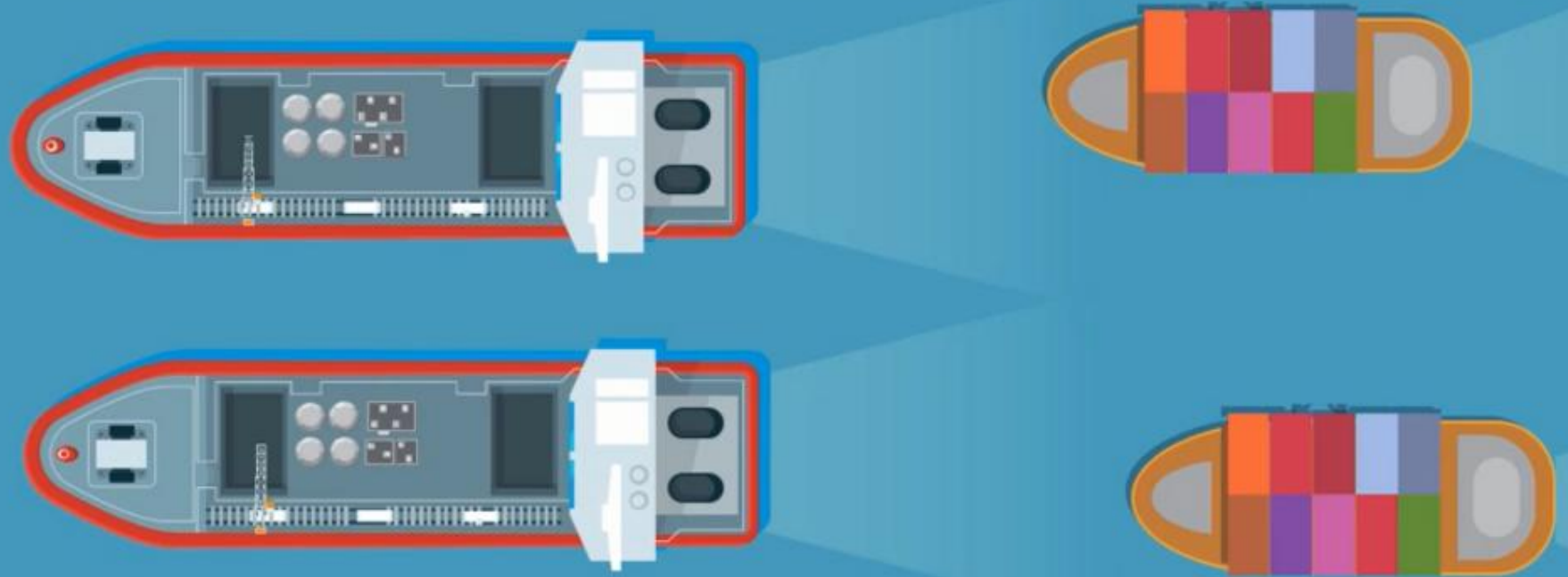
HA for Master Node



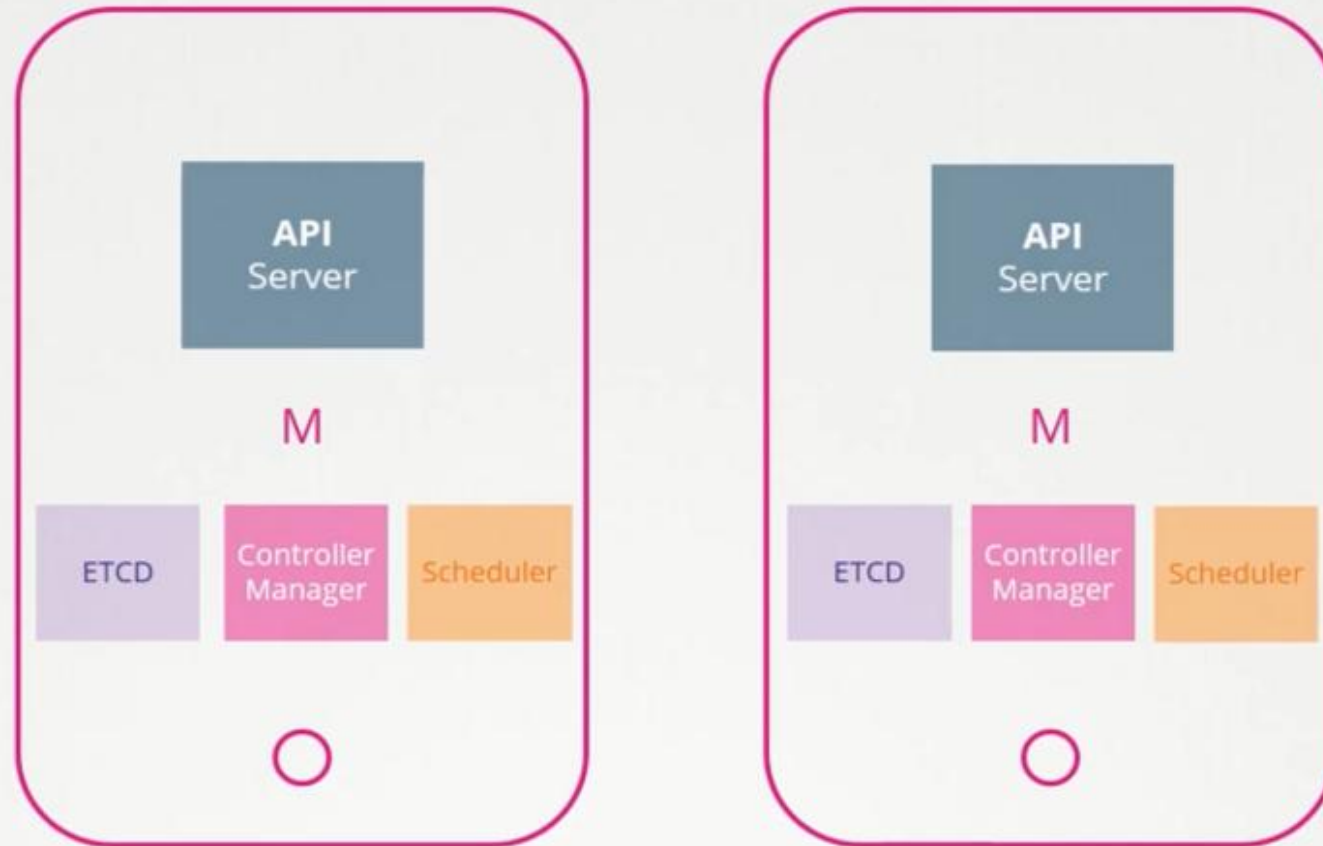
HA for Master Node



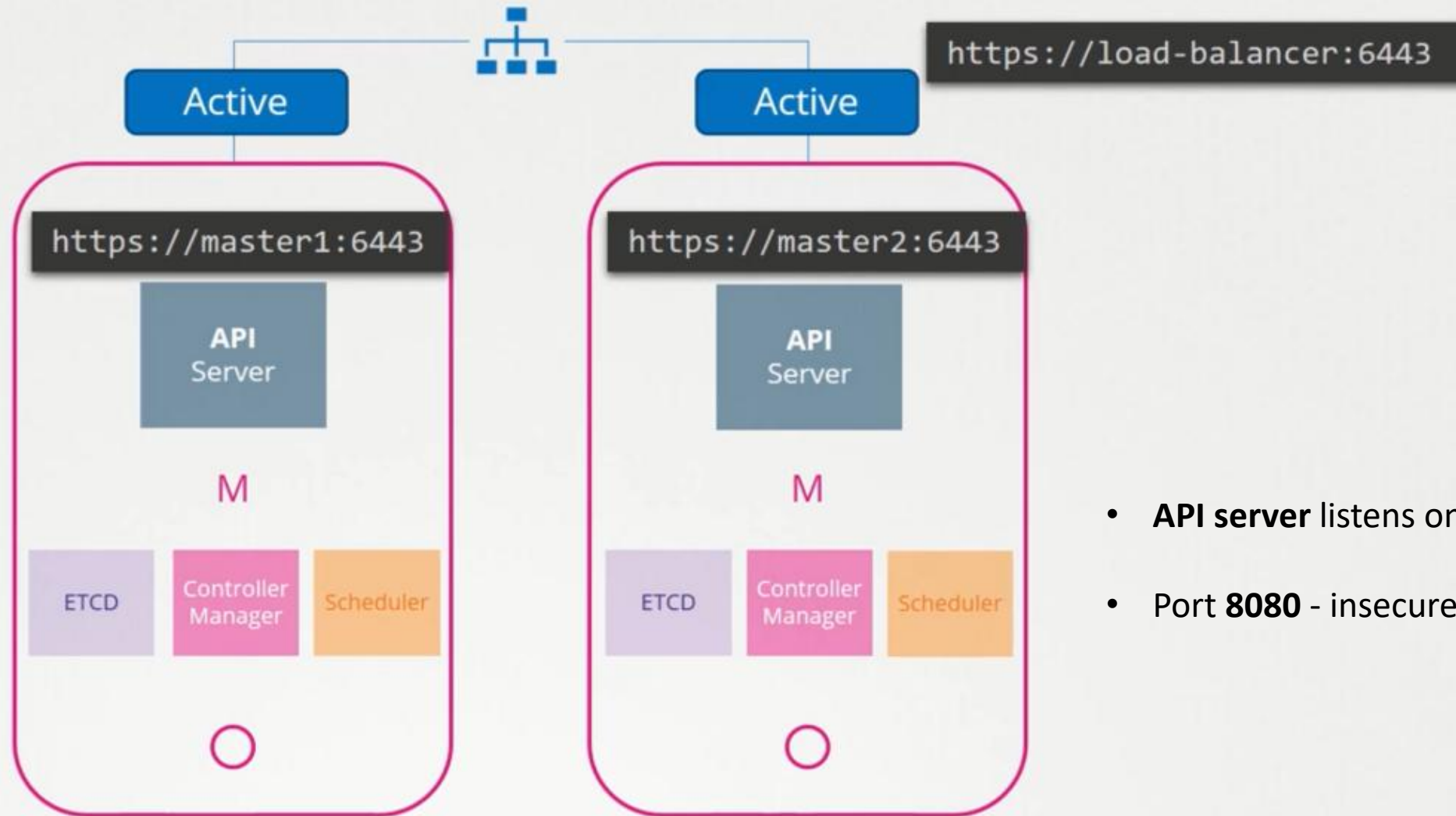
HA for Master Node



HA for Master Node

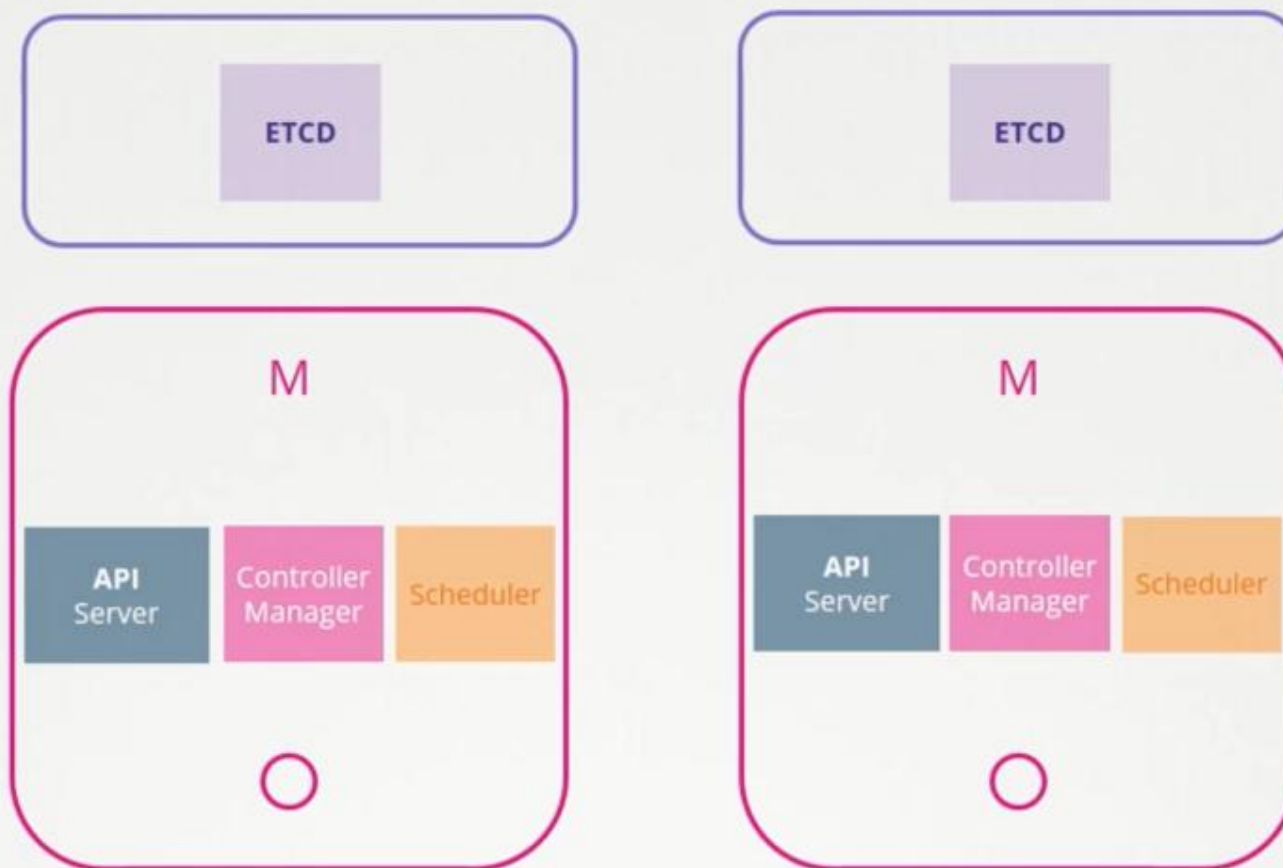


HA for Master Node



- **API server** listens on port **6443** (HTTPS)
- Port **8080** - insecure/localhost only

External ETCD Topology



- ✓ Less Risky
- ❖ Harder to Setup
- ❖ More Servers

Stacked Topology



- ✓ Easier to setup
- ✓ Easier to manage
- ✓ Fewer Servers
- ❖ Risk during failures

Controller



Master

Manage, Plan, Schedule, Monitor
Nodes

Controller-Manager

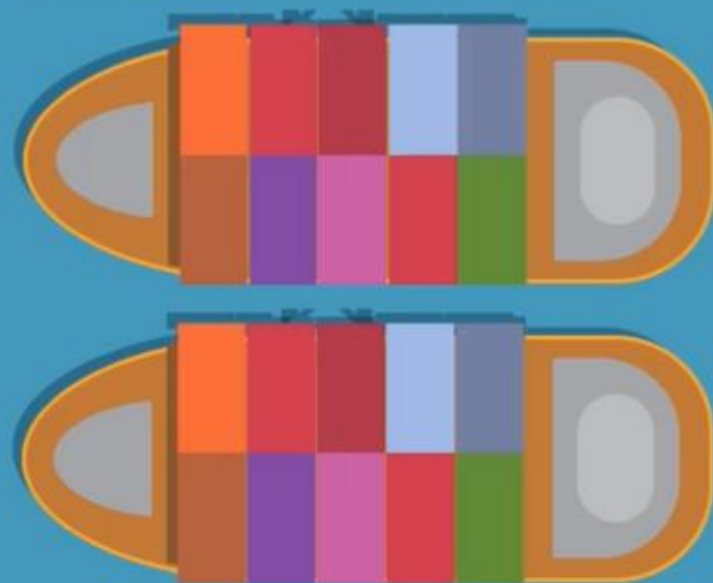
ETCD
CLUSTER

kube-scheduler



Worker Nodes

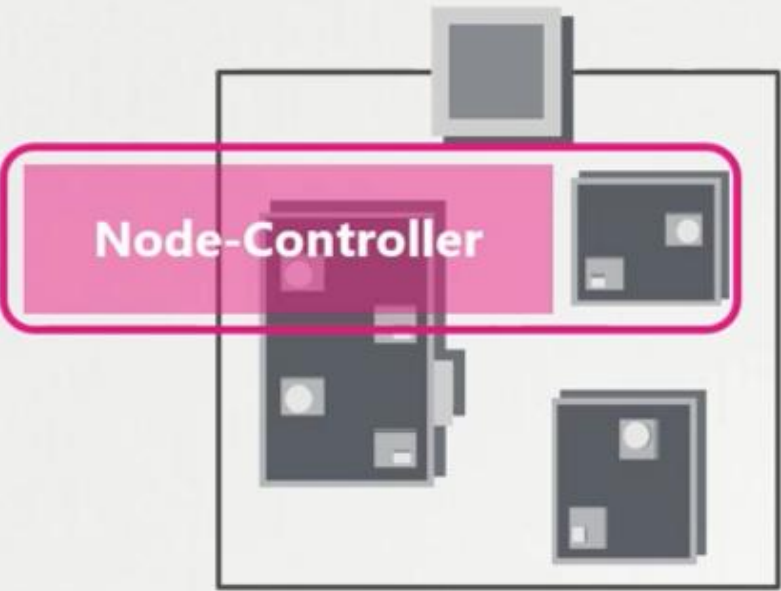
Host Application as Containers



I Controller

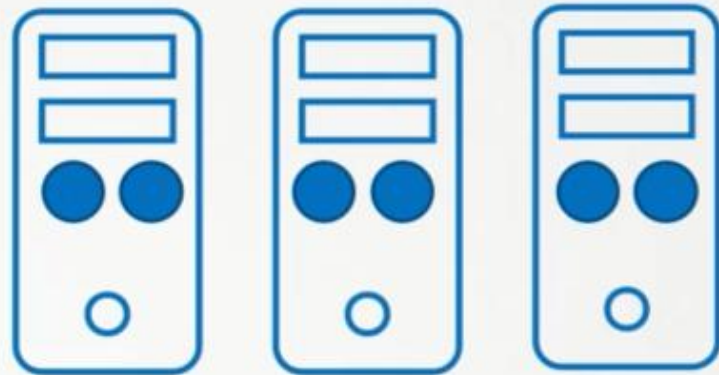
Watch Status

Remediate Situation



Node-Controller

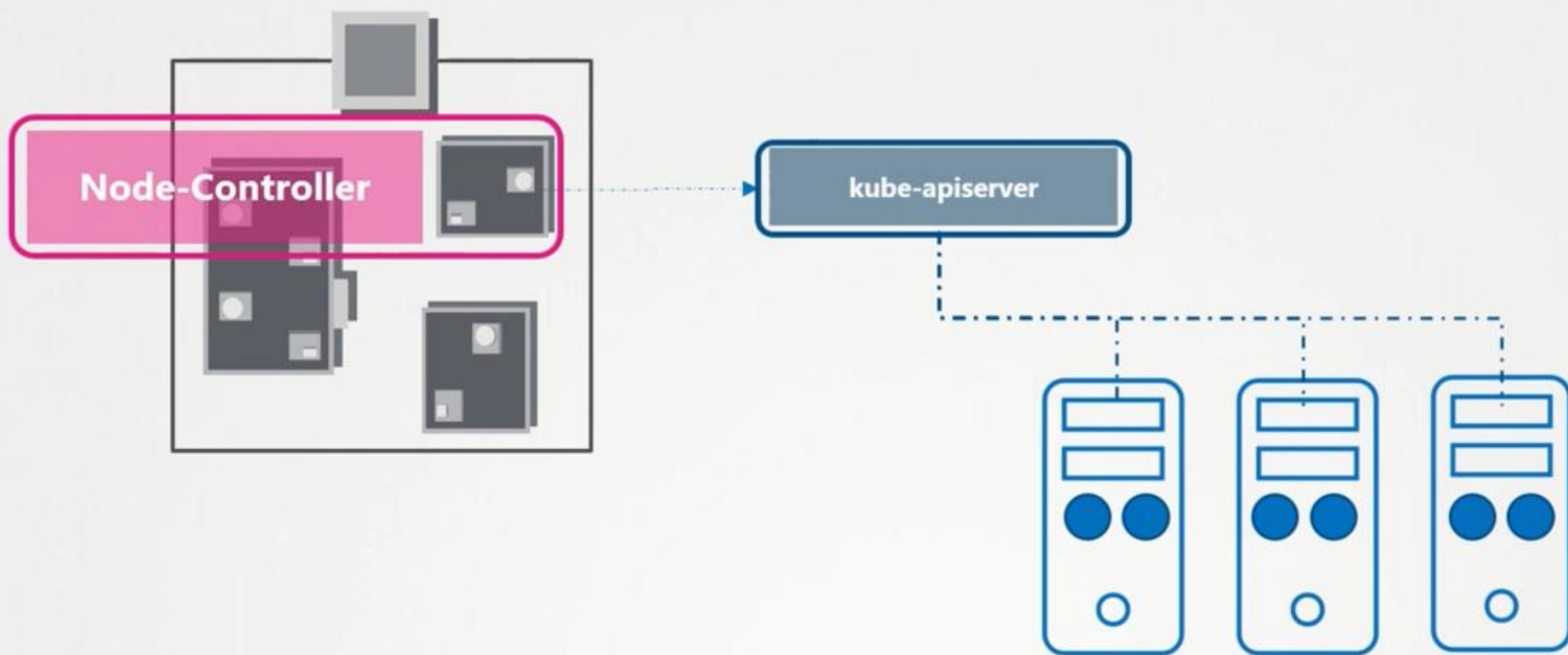
The diagram shows a central computer monitor icon at the top. Below it, a pink rounded rectangle with the text 'Node-Controller' is overlaid on a cluster of server rack icons. The entire cluster is enclosed in a black rectangular frame.



Controller

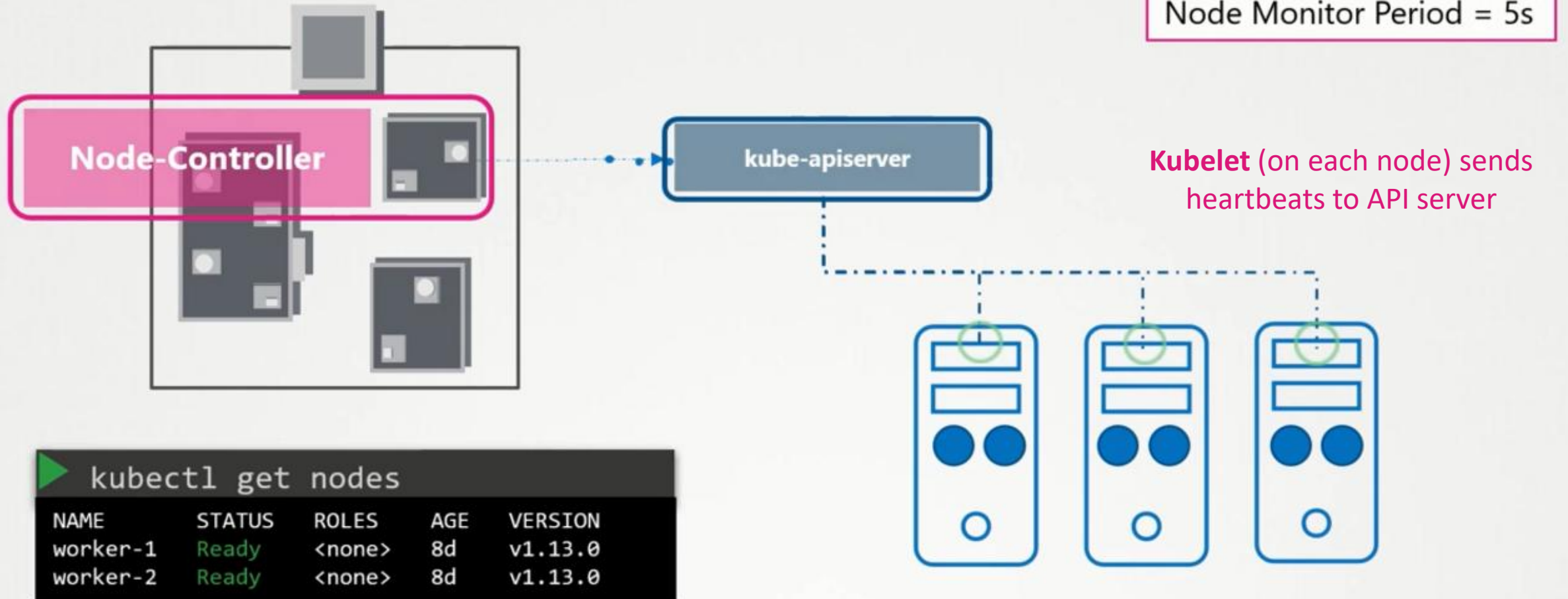
Watch Status

Remediate Situation

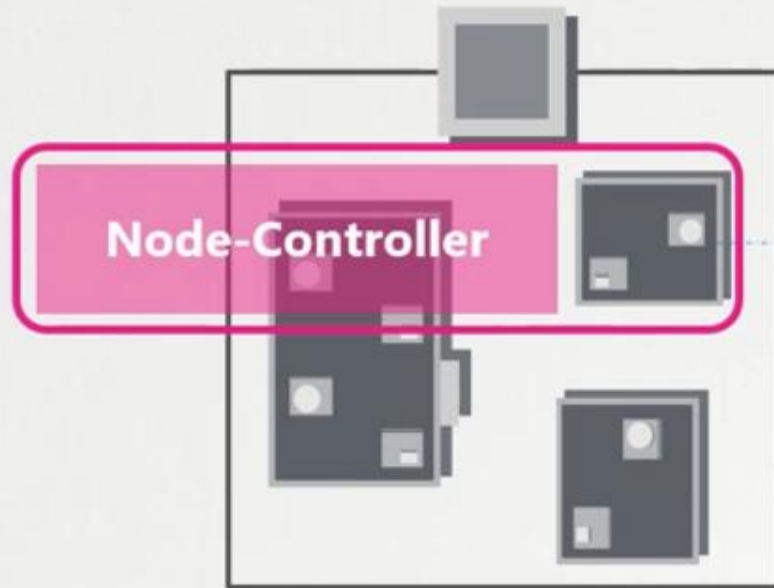


Controller

Controller manager watches these updates



Controller



kube-apiserver



UNREACHABLE

Watch Status

Remediate Situation

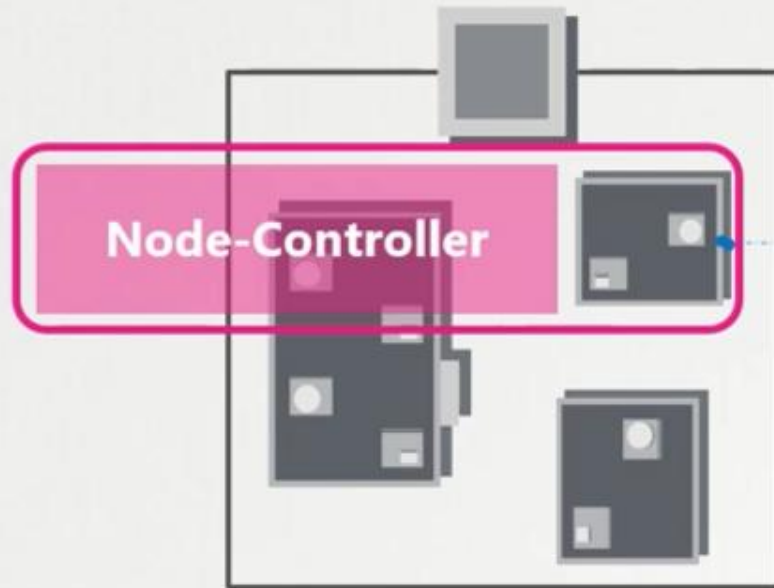
Node Monitor Period = 5s

Node Monitor Grace Period = 40s

```
► kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
worker-1	Ready	<none>	8d	v1.13.0
worker-2	NotReady	<none>	8d	v1.13.0

Controller



kube-apiserver



UNREACHABLE

Watch Status

Remediate Situation

Node Monitor Period = 5s

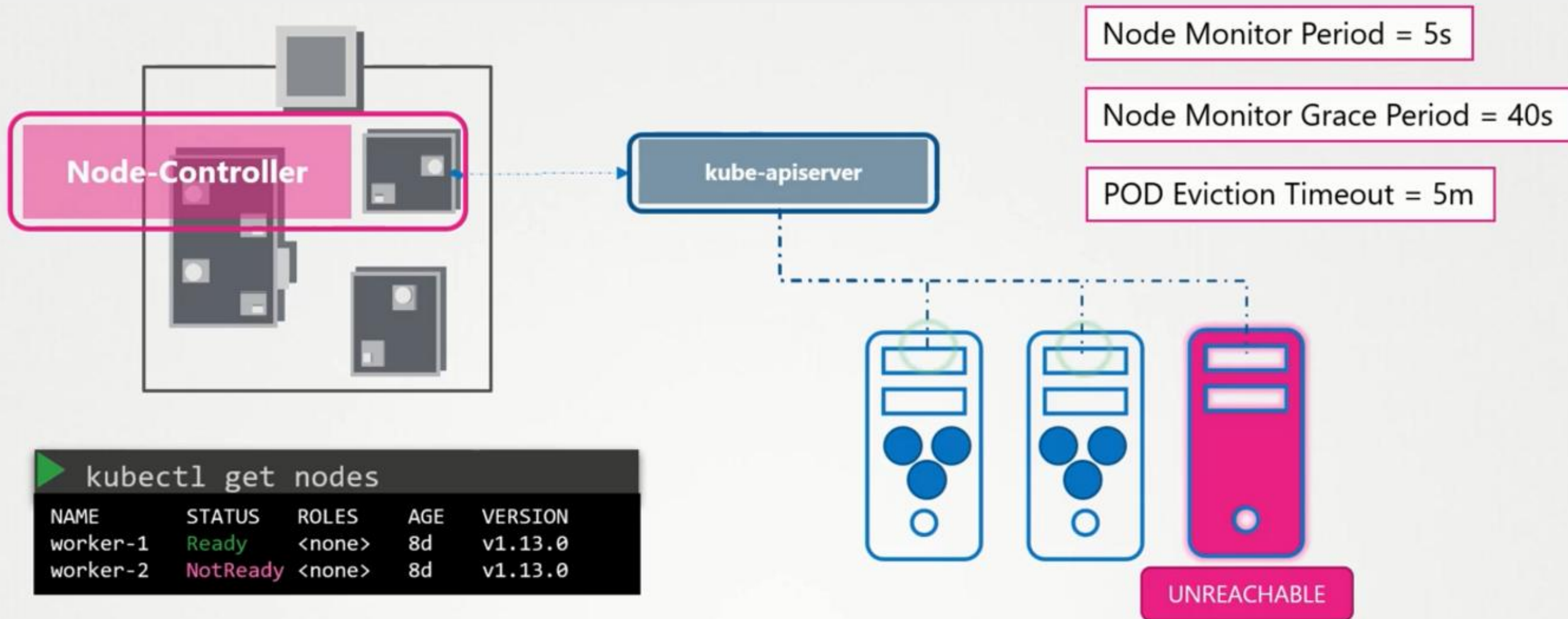
Node Monitor Grace Period = 40s

POD Eviction Timeout = 5m

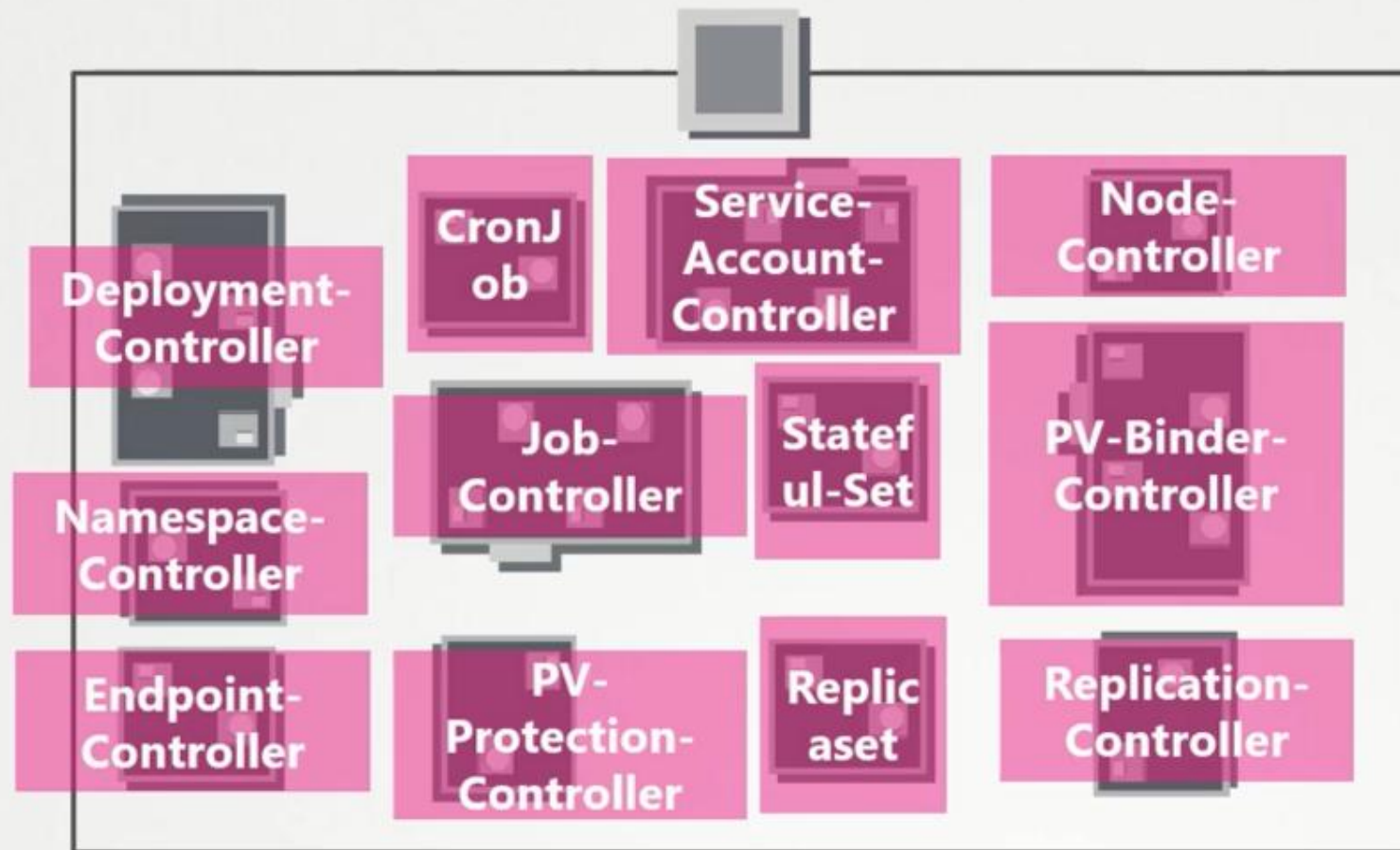
```
▶ kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
worker-1	Ready	<none>	8d	v1.13.0
worker-2	NotReady	<none>	8d	v1.13.0

- Controller Manager: "This pod is dead; we need a new one" → creates unscheduled pod
- Scheduler: "Best node for this pod is Node-2" → writes assignment to API server
- Kubelet on Node-2: "I see a pod assigned to me" → pulls image and starts containers



Controller



Watch Status

Remediate Situation

Node Monitor Period = 5s

Node Monitor Grace Period = 40s

POD Eviction Timeout = 5m

View kube-controller-manager - kubeadm

```
kubectl get pods -n kube-system
```

NAMESPACE	NAME	READY	STATUS	RESTARTS	AGE
kube-system	coredns-78fcd6f6894-hwrq9	1/1	Running	0	16m
kube-system	coredns-78fcd6f6894-rzhjr	1/1	Running	0	16m
kube-system	etcd-master	1/1	Running	0	15m
kube-system	kube-apiserver-master	1/1	Running	0	15m
kube-system	kube-controller-manager-master	1/1	Running	0	15m
kube-system	kube-proxy-lzt6f	1/1	Running	0	16m
kube-system	kube-proxy-zm5qd	1/1	Running	0	16m
kube-system	kube-scheduler-master	1/1	Running	0	15m
kube-system	weave-net-29z42	2/2	Running	1	16m
kube-system	weave-net-snm1	2/2	Running	1	16m

All Controllers bundled inside the **Controller Manager** component (one process running many controllers).

Kube Scheduler



Master

Manage, Plan, Schedule, Monitor
Nodes

Controller-
Manager

ETCD
CLUSTER

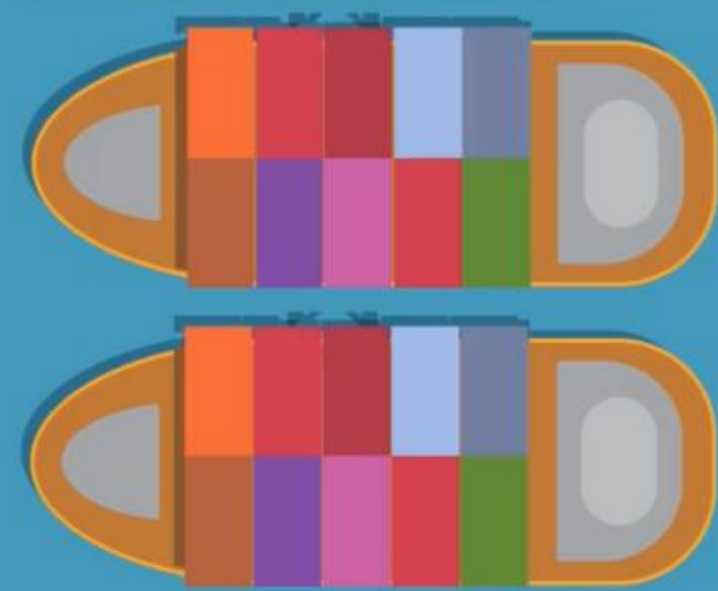


Kube-Scheduler

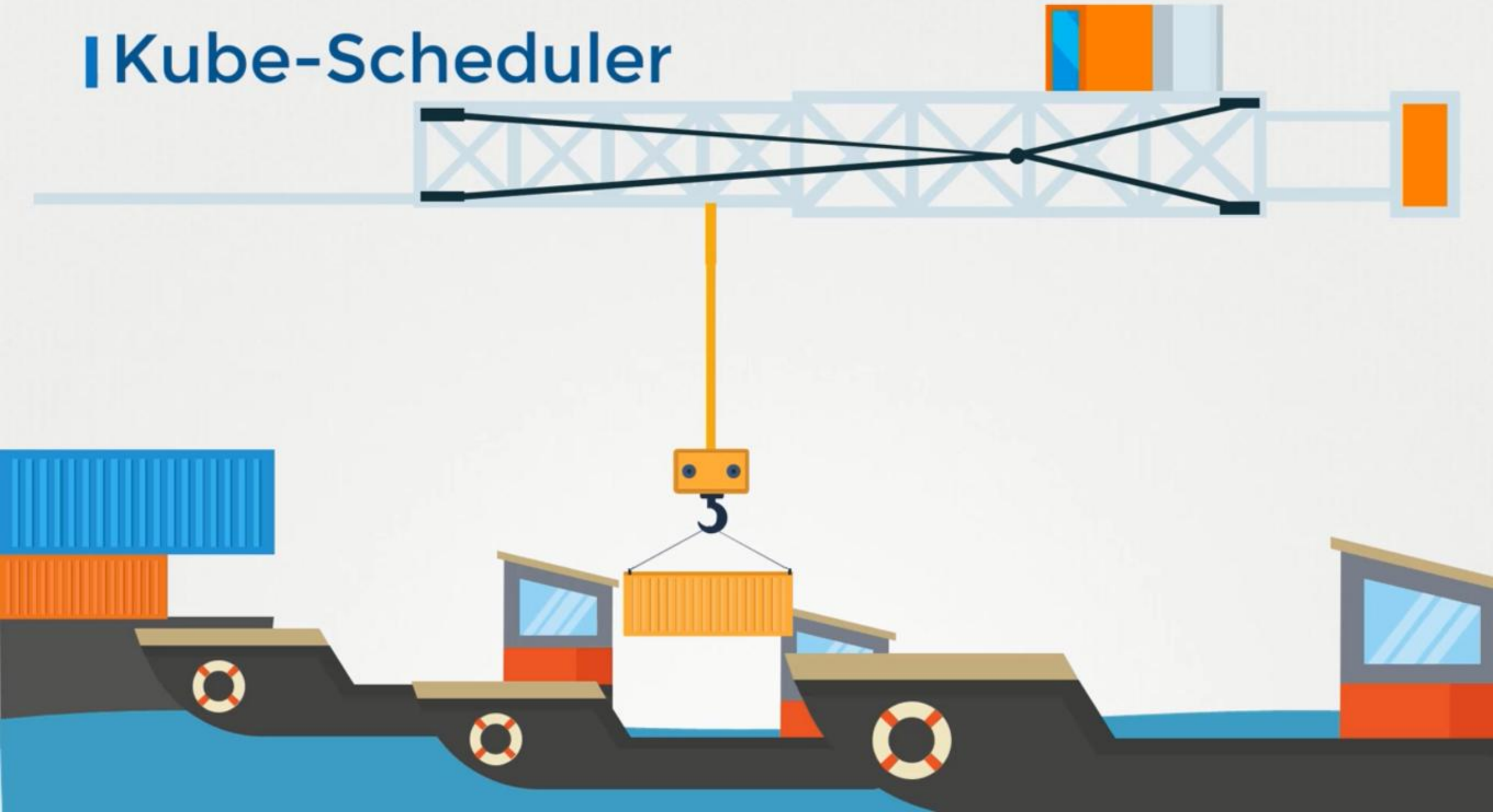


Worker Nodes

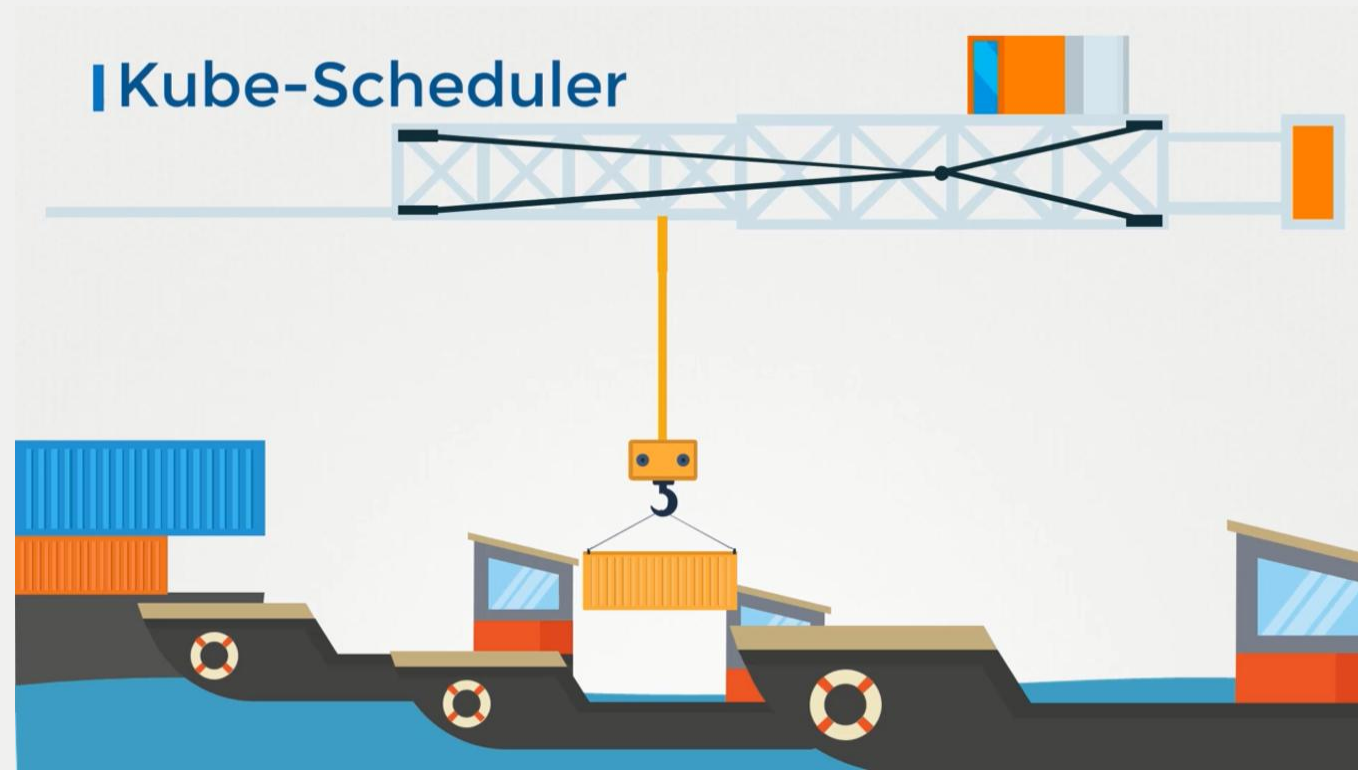
Host Application as Containers



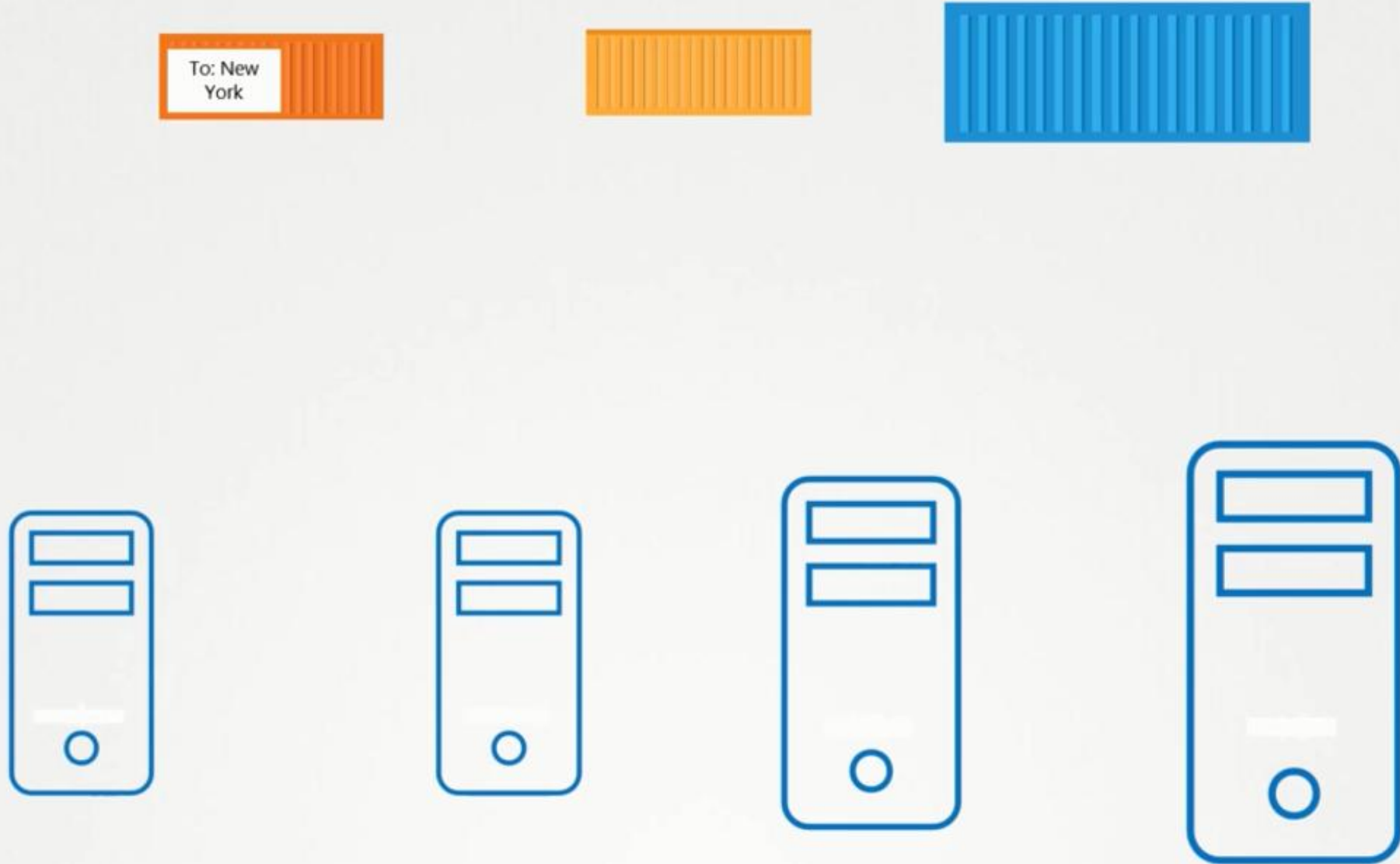
Kube-Scheduler



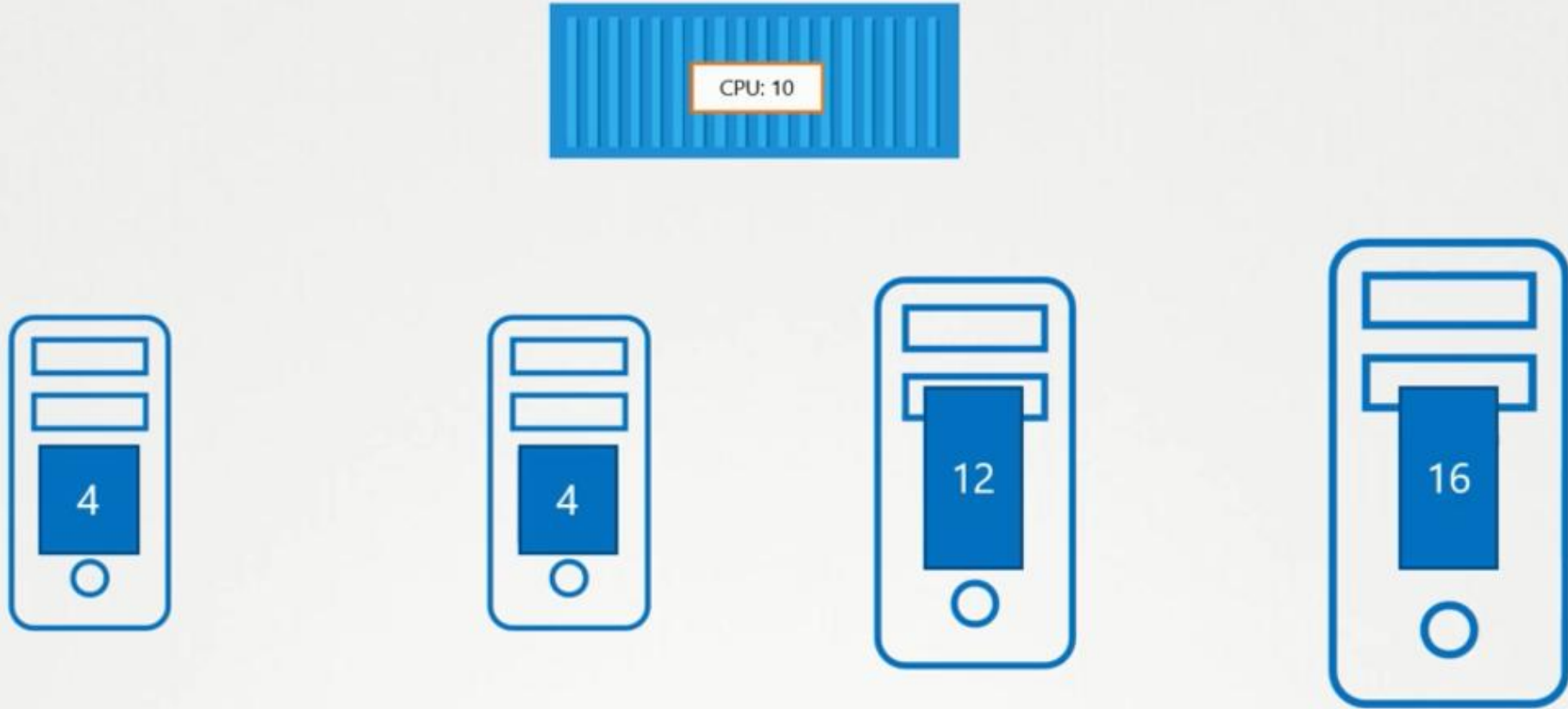
- Watches for new pods that don't have a home yet
 - Finds the best node to run them on (checks CPU, memory, rules)
 - Tells the API server "put this pod on that node": (HOW)
1. Scheduler watches the API server for unscheduled pods
 2. Scheduler decides which node is best
 3. Scheduler updates the pod's information in the API server (writes the node assignment)



| Kube-Scheduler

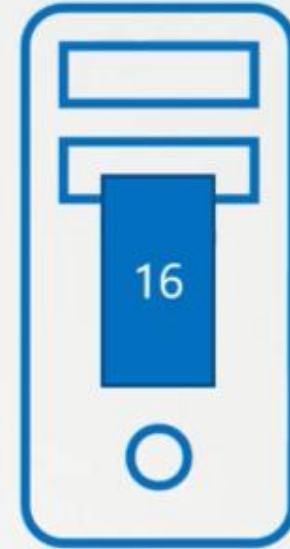
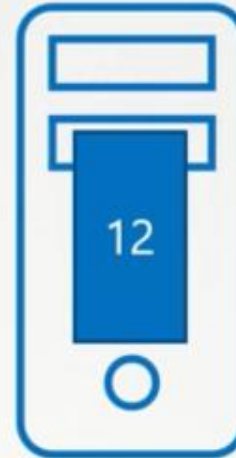
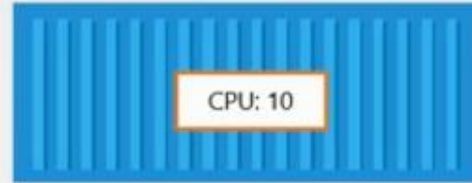


| Kube-Scheduler

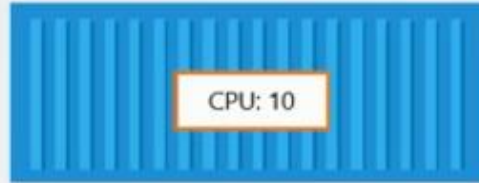


Kube-Scheduler

1. Filter Nodes

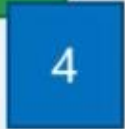


Kube-Scheduler



1. Filter Nodes

2. Rank Nodes



Now let's go to see
Different Parts in a **Worker Node** of k8s